# Probabilistic Models of the Visual Cortex

# Fall 2026 Homework 2

Prof. Alan Yuille

## Instructions

1. **Deadline**: Monday, Nov 2, 2026 at 11:59 PM (Eastern Time).
2. **Late homework**: Late homework will not be accepted unless permission is obtained well in advance for documented extenuating circumstances. Email Prof. Yuille for approval and copy the TAs.
3. **Submission**: one PDF file on Gradescope with your written answers and the plots and answers for Questions 9–11, named `<your_jhed>-hw2.pdf`. Code is not required and will not be graded; a notebook export that shows code is fine (File → Save and Export Notebook As → HTML, then print it to PDF from your browser).
4. **Coverage**: Lectures 9–14 (Questions 1–2 Lecture 9; Question 3 Lecture 10; Question 4 Lecture 11; Questions 5–6 Lecture 12; Question 7 Lecture 13; Question 8 Lecture 14). 110 points: written Questions 1–8 (72), experimental Questions 9–11 (38).
5. **Warning**: Start the experimental section (Questions 9–11, three notebooks) early! Do not use AI tools (such as ChatGPT, Claude, Gemini or Copilot) to complete this homework. If we detect that AI was used to complete it, you will receive zero points for this homework.
6. **Calculations**: You may use a calculator or Python. Write down the formula and the numbers you plug in, not only the final answer. Give four significant digits; log is the natural logarithm.
7. **Collaboration**: Feel free to discuss the questions with friends, but be sure to write your answers entirely independently!
8. **Questions?** Attend office hours (see the course website) or send an email!

# Question 1: The Bayes-Kalman filter and physical world models (11 points)

1. **(3 points) Concepts.**

   (a) What are the two steps of the Bayes-Kalman filter, and what are their equations? **(1 point)**

   (b) What is the purpose of the second step? Derive its equation from Bayes' rule, conditioning on $Y_t = \{y_1, \ldots, y_t\}$ throughout, and mark where the conditional independence of $y_{t+1}$ and $Y_t$ given $x_{t+1}$ is used. **(2 points)**

2. **(1 point) Figure 1: one step of the filter.** Figure 1 shows the filter at work: the red circle, then the blue, then the green.

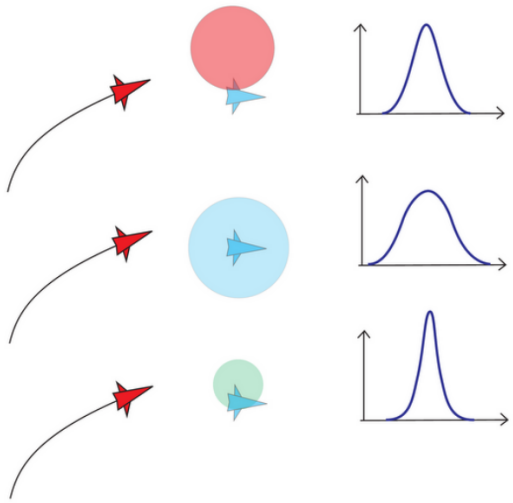


*Figure 1: Three successive distributions of the airplane's position.*

Why is the blue circle wider than the red one and the green circle narrower than the blue one?

3. **(4 points) Two steps of the filter by hand.** A moving object follows the Gaussian model $P(x_{t+1} \mid x_t) = \mathcal{N}(x_t + \mu, \sigma_p^2)$ with $\mu = 1$ and $\sigma_p^2 = 1$; $P(y_t \mid x_t) = \mathcal{N}(x_t, \sigma_m^2)$ with $\sigma_m^2 = 4$; $P(x_1) = \mathcal{N}(\mu_0, \sigma_0^2) = \mathcal{N}(0, 3^2)$; the filter keeps $P(x_t \mid Y_t) = \mathcal{N}(\mu_t, \sigma_t^2)$. Observations: $y_1 = 1.0$, $y_2 = 3.5$.

   (a) Initialise the filter with $y_1$ — the correction step with the prior $P(x_1)$ in place of the prediction — and give $P(x_1 \mid y_1) = \mathcal{N}(\mu_1, \sigma_1^2)$ and the prediction $P(x_2 \mid y_1)$ (mean and variance). **(1 point)**

   (b) Correct with $y_2$: give the Kalman gain $K_2$, the corrected mean $\mu_2$ and the variance $\sigma_2^2$. Then write $\mu_2$ as a weighted average of the prediction $\mu + \mu_1$ and the observation $y_2$, with the two weights in symbols and as numbers (the linear rule of Lecture 5 for combining two cues). **(2 points)**

   (c) Compute the fixed point $\sigma^2$ of the variance sequence $\sigma_1^2, \sigma_2^2, \ldots$ and the gain $K$ at the fixed point. **(1 point)**

4. **(3 points) Motion perception and the physical world.**

   (a) In a random-dot display 38 dots move at random and one along a straight line; after a few frames that dot is seen, although in a single frame every dot looks alike. Explain why in terms of prediction and correction. **(1 point)**

   (b) In Wu's physical world model a physics engine takes the world state and the action to the next world state, and a graphics engine renders each world state into an image. Which is the motion model and which the observation model of the Bayes-Kalman filter? **(1 point)**

   (c) What is inverse graphics? **(1 point)**

# Question 2: Dynamic programming, Markov models and hidden Markov models (10 points)

1. **(3 points) Concepts.**

   (a) What are the three basic problems of a hidden Markov model, and which algorithm solves each? **(1 point)**

   (b) In what sense is the recursion $\alpha_{t+1}(j) = \left[\sum_i \alpha_t(i) a_{ij}\right] b_j(O_{t+1})$ the forward pass of dynamic programming: which two operations are exchanged? **(1 point)**

   (c) Of which algorithm of Lecture 8 is the learning algorithm of (a) a case? **(1 point)**

2. **(3 points) Dynamic programming.** In Figure 2 the number in each circle is $h_m(x_m)$, the lowest cost of reaching that value, and the teal edge into a circle is the kept best way into it.

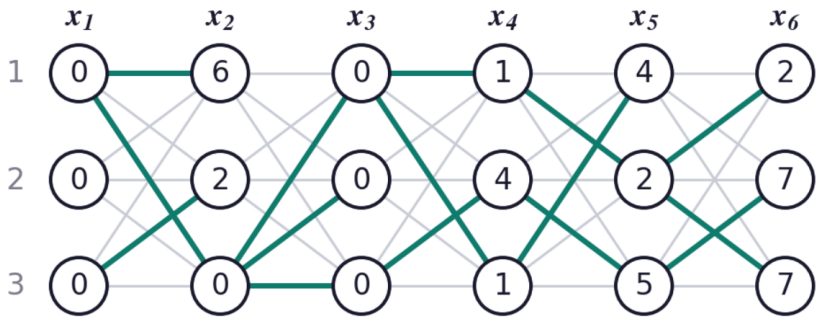



*Figure 2: The forward pass on a chain of six variables with three values each.*

   (a) From Figure 2 alone, give the lowest total cost and the configuration $(\hat{x}_1, \ldots, \hat{x}_6)$: start the backward pass at the value of $x_6$ with the smallest $h_6$ and follow the teal edges back. **(1 point)**

(b) A friend who claims psychic powers ($x_3$) predicts that a coin will land heads ($x_1$), which a two-headed coin ($x_4$) would also explain; a levitation test ($x_2$) depends on the psychic powers only; all four variables are binary. Write $P(x_1, x_2, x_3, x_4)$ as a product of one conditional distribution per node. How many terms remain when $P(x_1 = 1)$ is computed with the leaf $x_2$ summed out first, against $2^3 = 8$ for the direct sum over $(x_2, x_3, x_4)$? **(2 points)**

3. **(4 points) An observable Markov model is learnt by counting.** Sadie suffers from allergy attacks during the autumn. Her record over 30 days (1 = attack, 0 = no attack) is

```
0 0 0 1 1 0 0 0 0 0 1 1 1 0 0 0 0 1 0 0 0 0 0 0 1 1 0 0 0 0
```

Treat it as one sequence from an observable Markov model with two states, $s_1$ = no attack and $s_2$ = attack.

(a) Define a first-order Markov model: the Markov property and the two sets of numbers, with their constraints, that specify it. **(1 point)**

(b) Count the transitions $n_{ij}$ from $s_i$ to $s_j$ in the record (row $i$ = from, column $j$ = to) and give the table of estimated transition probabilities $\hat{a}_{ij} = n_{ij} / \sum_j n_{ij}$. Why are the counts all that the maximum likelihood estimate needs? **(2 points)**

(c) What is the estimated probability of an attack tomorrow if Sadie is healthy today, and if she has an attack today? **(1 point)**

# Question 3: Regression, perceptrons, backpropagation and convolutional networks (12 points)

1. **(3 points) Concepts.**

   (a) What is regression in the sense of this lecture: what is learnt, from what, and by which criterion? Why can a network trained this way fail on images from a different data domain? **(2 points)**

   (b) State the perceptron's classification rule and the update of the perceptron algorithm when an example is misclassified. **(1 point)**

2. **(4 points) Figure 3: the disc and the ring.** Figure 3 shows points in a disc (red, $y = +1$) and in a ring around it (blue, $y = -1$), with the line of logistic regression (panel 1), the three lines on which the hidden units of a trained three-unit multi-layer perceptron (MLP) equal 1/2 (panel 2) and the curve where the MLP's output is 1/2 (panel 3).

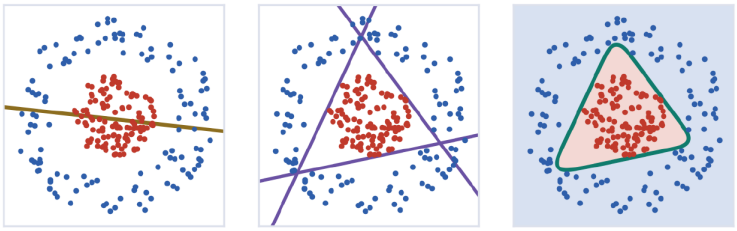


*Figure 3: The disc, the ring and the three panels.*

   (a) Panel 1: why can logistic regression not separate the disc from the ring, whatever its weights? Panel 3: inside the disc two of the hidden units are close to 1 and one is close to 0; why is the region where the output exceeds 1/2 a triangle? **(2 points)**

(b) **Backpropagation by hand.** The MLP has hidden units $z_i = \sigma(u_i)$, $u_i = \vec{\Omega}^i \cdot \vec{x} + \Omega_0^i$, $i = 1, 2, 3$, with $\sigma$ the sigmoid, output $y = \sigma(u)$, $u = \vec{w} \cdot \vec{z} + w_0$, and energy $E = -t \log y - (1 - t) \log(1 - y)$ for an example with label $t$. Take $\vec{x} = (2, 0)$, $t = 0$, $\vec{\Omega}^1 = (1, 0)$, $\Omega_0^1 = -0.5$; $\vec{\Omega}^2 = (0, 1)$, $\Omega_0^2 = 0.5$; $\vec{\Omega}^3 = (-1, -1)$, $\Omega_0^3 = 1$; $\vec{w} = (1, -2, 1)$, $w_0 = 0.5$. (i) Forward pass: compute $u_i$, $z_i$, $u$, $y$ and $E$. (ii) Backward pass, with $\partial E/\partial u = y - t$ and $\delta_i = \partial E/\partial u_i = (y - t)\, w_i\, z_i(1 - z_i)$: compute $\partial E/\partial w_i$, $\partial E/\partial w_0$, $\delta_1, \delta_2, \delta_3$, $\partial E/\partial \vec{\Omega}^2$ and $\partial E/\partial \Omega_0^2$. **(2 points)**

3. **(3 points) AlexNet.**

(a) State the two restrictions that turn a multi-layer perceptron into a convolutional neural network. **(1 point)**

(b) What are the two effects of max pooling named in the lecture? **(1 point)**

(c) AlexNet uses two methods to reduce overfitting: name them and say what each does at training time. **(1 point)**

4. **(2 points) Fine-detail tasks.**

(a) In its loss, holistically-nested edge detection (HED) weights edge pixels by $\beta = |Y_-|/|Y|$, the fraction of non-edge pixels, and non-edge pixels by $1 - \beta$. If 8% of the pixels are edges, what is $\beta$, and how many false alarms does one missed edge cost? **(1 point)**

(b) A residual block computes $H_{l+1} = H_l + F(H_l, W_l)$. Write its backward pass $\partial L/\partial H_l$ in terms of $\partial L/\partial H_{l+1}$: which term lets the error reach the early layers of a 40-layer network, where a plain 40-layer network fails? **(1 point)**

# Question 4: Adversarial examples and defenses (8 points)

1. **(3 points) Concepts.**

   (a) With $S = \{r : |r_i| \leq \varepsilon\}$ the set of allowed perturbations $r$ of an image $x$ ($\varepsilon$ per pixel), write the fast gradient sign method (FGSM) and one step of projected gradient descent (PGD), with what the projection $\Pi_{x+S}$ does to each pixel. **(1 point)**

   (b) What is the difference between a white-box attack and a blind (black-box) attack? **(1 point)**

   (c) Why is PatchAttack a black-box attack? **(1 point)**

2. **(2 points) The linear argument by hand.** A linear unit computes $w^\top x$ with $w = (2, -1, 0.5, -3, 1.5)$, and each input may change by at most $\varepsilon = 0.5$. (i) Which perturbation $\eta$ raises $w^\top(x+\eta)$ most, and by how much (general formula and number)? (ii) For random signs $\eta_i = \pm 0.5$, independent and equally likely, what are the mean and the standard deviation of $w^\top \eta$, and how many times smaller is this standard deviation than the change in (i)?

3. **(2 points) Figure 4: standard training and min-max training.**

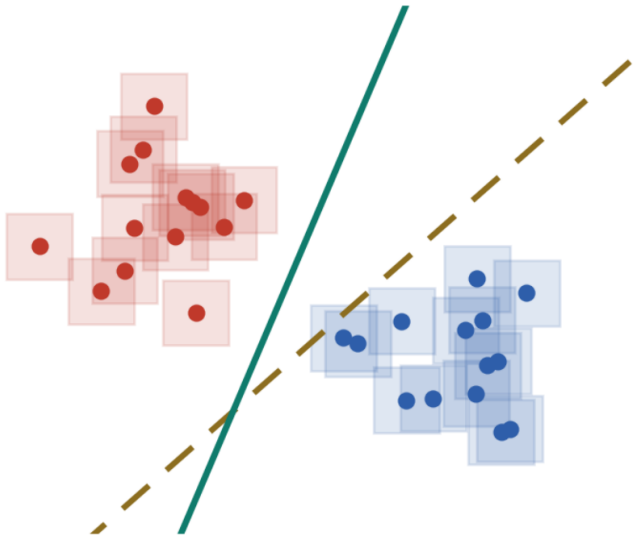


*Figure 4: Two classes of points, red and blue, each with a square of half-side $\varepsilon$ around it, and two separating lines, dashed olive and solid teal: one comes from standard training, the other from min-max training.*

One of the two lines cuts some of the squares, the other none. (i) Write the two optimisation problems that the lecture places side by side, standard training of the weights $\theta$ and the attack by a perturbation $r \in S$, and the min-max training objective that combines them. (ii) Which line in Figure 4 is the perceptron of Lecture 10 (standard training), and why?

4. **(1 point) Visual concepts.** What is a visual concept: what is clustered, and with which algorithm of Lecture 8?

# Question 5: Compositional networks (8 points)

1. **(3 points) Concepts.**

   (a) What are synthesis and analysis in the formulation $P(W \mid I) = P(I \mid W)\, P(W)/P(I)$, and in what sense is a CompNet only an *approximate* analysis by synthesis? **(2 points)**

   (b) Write the von Mises-Fisher (vMF) kernel $p(f_p \mid \lambda_k)$, $\lambda_k = \{\mu_k, \sigma_k\}$, of a unit feature vector $f_p$, and say what $\mu_k$ and $\sigma_k$ are. **(1 point)**

2. **(2 points) Figure 5: the CompNet with the outlier model.** In Figure 5 the red box adds the outlier model with its five occluder models $\beta_1, \ldots, \beta_5$.

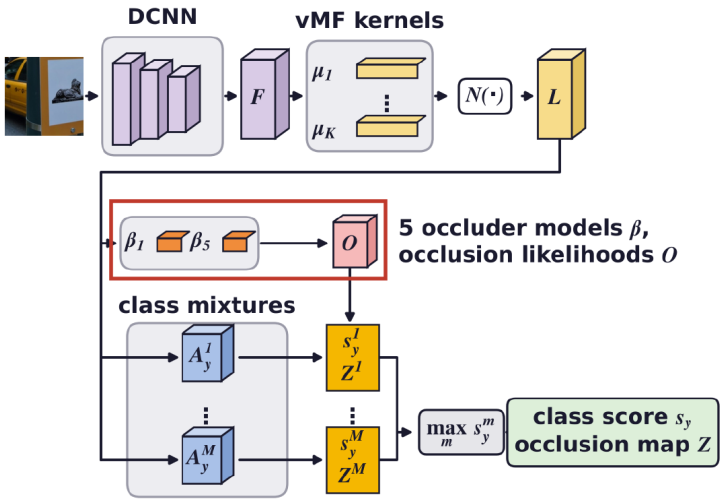



*Figure 5: The CompNet as a feed-forward network with the outlier model.*

(i) Trace the forward pass: what is $F$, what is the block $L$ (with its size), and which operation produces $L$ from $F$? (ii) Why are the occluder models the same at every position?

3. **(2 points) The competition at one position.** The outlier model $p(f_p \mid \beta, \Lambda) = \sum_k \beta_k\, p(f_p \mid \lambda_k)$ shares the object's kernels $\Lambda = \{\lambda_k\}$; $z_p^m = 1$ says that the occluder generated $f_p$, with prior $\pi = p(z_p^m = 1)$. (i) Write the decision that sets $z_p^m = 1$ as a log-likelihood-ratio test of the occluder against the object mixture, with its threshold. (ii) At one position $\log p(f_p \mid \beta, \Lambda) = -2.5$, the object mixture $A_{p,y}^m$ of the best component gives $\log p(f_p \mid A_{p,y}^m, \Lambda) = -6.0$, and

$\pi = 0.2$: is the position declared occluded, and what does it contribute to the component score $\sum_p \max\{\log p(f_p, z_p^m{=}0),\ \log p(f_p, z_p^m{=}1)\}$?

4. **(1 point) Patch attacks.** Why are CompNets robust to patch attacks "out of the box", without adversarial training?

# Question 6: Compositional models and unsupervised structure learning (7 points)

1. **(2 points) Concepts.** (i) Write the exponential model $P(\mathbf{W} \mid \mathbf{I})$ of a hierarchical compositional model with its three kinds of potential, and say what question each kind answers. (ii) The lecture distinguishes three tasks, inference, learning and structure induction (the task of the PGMM and of the horse dictionary): what is assumed known in each?

2. **(1 point) Figure 6: part sharing.** In Figure 6 squares are AND nodes, circles elementary models and the orange triangle an OR node.

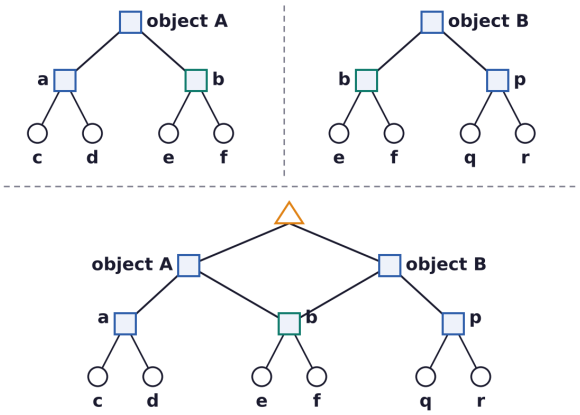



*Figure 6: Part sharing between two objects.*

Count the nodes (squares, circles and the triangle) in the two separate models of objects A and B together (top row) and in the shared graph (bottom row).

3. **(2 points) AND/OR graphs.** (i) What are AND nodes and OR nodes, and what does the switch variable $t_\nu$ of an OR node do? (ii) Why can a graph with AND nodes only, like the horse model of the lecture, describe one pose and one viewpoint only?

4. **(2 points) Learning the structure.** The hierarchical dictionary learns each level bottom-up from the level below in four steps: composition, clustering, suspicious coincidence and competitive exclusion. (i) What does each step do? (ii) Why is the third step, which keeps a concept only if it occurs in at least 90 % of the training images, "the key test"?

# Question 7: Few-shot and self-supervised learning (8 points)

1. **(2 points) Concepts.**

   (a) What is a proxy task? **(1 point)**

   (b) In the few-shot method of the lecture, what does the parameter predictor $\phi$ compute for a new category, and from what? **(1 point)**

2. **(1 point) Mean activations and weights.** Figure 7 maps the categories by their mean activations $\bar{\mathbf{a}}$ (left) and by their weights $w$ (right).

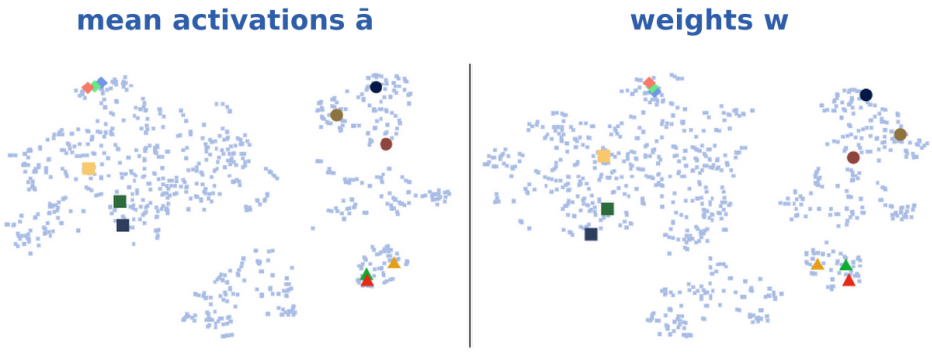



*Figure 7: Two t-SNE maps of the categories of a ResNet-50.*

What conclusion does the lecture draw from where the four groups of categories marked by diamonds, circles, squares and triangles lie in the two maps?

3. **(3 points) The contrastive loss.** A batch holds $n = 2$ images, so $2n = 4$ views, embedded as unit vectors: $\mathbf{z}_1, \mathbf{z}_2$ from image A and $\mathbf{z}_3, \mathbf{z}_4$ from image B, with $\mathbf{z}_1 \cdot \mathbf{z}_2 = 0.866$, $\mathbf{z}_1 \cdot \mathbf{z}_3 = 0$ and $\mathbf{z}_1 \cdot \mathbf{z}_4 = -0.5$; each image is its own class.

   (a) Write the contrastive loss of the lecture and define its symbols. Then, with temperature $\tau = 0.5$, compute the loss term of query $\mathbf{z}_1$: the three exponentials, the probability the softmax assigns to the positive key $\mathbf{z}_2$, and $-\log$ of it. **(2 points)**

   (b) DINO and iBOT use no negative keys, so the trivial solution in which the network gives the same output for every image (collapse) would satisfy their self-distillation loss. Which two operations on the teacher's output prevent it? **(1 point)**

4. **(2 points) Masked image modeling.**

   (a) BEiT and iBOT both predict something about the masked patches. What does each predict for a masked patch? **(1 point)**

   (b) MaskFeat regresses the histogram of oriented gradients (HOG) of the masked patches instead of their pixel colours. Why are pixel colours a poor target and HOG a better one? **(1 point)**

# Question 8: Transformers: attention as a weighted average and as clustering (8 points)

1. **(2 points) Concepts.**

   (a) Convolution and self-attention both compute a weighted average $y_o = \sum_p w_{op} x_p$ of the input feature vectors $x_p$. What is $w_{op}$ in each case? **(1 point)**

   (b) Why does a transformer need a positional encoding? **(1 point)**

2. **(1 point) Mask transformers.** A mask transformer outputs $N$ masks, each with a class distribution that includes "no object" ($\varnothing$). How are the $N$ predictions matched to the $K < N$ ground-truth masks during training?

3. **(2 points) Attention by hand.** Three tokens with $d = 2$: $x_1 = (1, 0)$, $x_2 = (0, 1)$, $x_3 = (1, 1)$; $W^Q = W^K = W^V = I$, so $q_p = k_p = v_p = x_p$. (i) Write scaled dot-product attention, defining the queries, keys and values. (ii) Compute the weights $a_{1p}$ of token 1 and the output $y_1 = \sum_p a_{1p} v_p$.

4. **(3 points) Figure 8: the same affinities normalised two ways.** In Figure 8 the weights are $e^{-|x_p - c_g|^2/2\sigma^2}$, with $x_p$ the colour of pixel $p$ and $c_g$ one of the three centres.

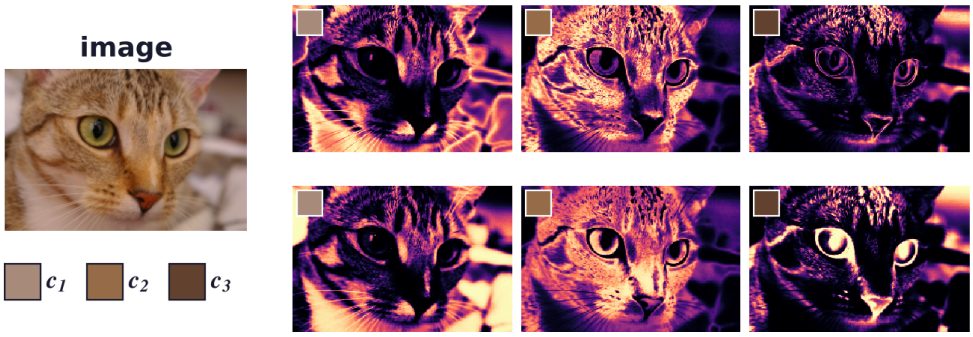



*Figure 8: Gaussian weights of the pixel colours for three k-means colour centres, normalised in two ways.*

(a) Which row has the form of the mask prediction $\hat{m}_g(p)$ of a mask transformer (a softmax over the clusters) and which the form of the query update $a_{gp}$ of cross-attention (a softmax over the pixels), and how can you tell from the maps? **(1 point)**

(b) Two centres and four pixels have the affinity logits $\ell_{gp}$: $(3, 1, 0, 0)$ for $g = 1$ and $(2, 2, 2, 2)$ for $g = 2$. Compute both normalisations of (a) for these logits, the softmax over the pixels and the softmax over the clusters. **(1 point)**

(c) Binary assignments $V_{pg}$ of pixels $p$ to clusters $g$ have the energy $E(V, c) = -\sum_{p,g} V_{pg}\, x_p^T c_g$ and $P(V) \propto e^{-E(V,c)}$. If each cluster takes exactly one pixel ($\sum_p V_{pg} = 1$ for every $g$), the pixel chosen by cluster $g$ has the expected feature $\sum_p \langle V_{pg} \rangle\, x_p$. Compare it with the cross-attention update $c_g \leftarrow c_g + \sum_p a_{gp} v_p$, $a_{gp} = e^{q_g^T k_p} / \sum_{p'} e^{q_g^T k_{p'}}$: with which projections $W^Q$, $W^K$, $W^V$ does the sum agree with the expected feature, and which term of the update does the expected feature lack? **(1 point)**

# Questions 9–11: Experimental Section (38 points)

In this section you will run three Jupyter notebooks. They come with this PDF in the zip file of the homework, together with the data. Unzip the file and keep the folder structure, so that the folder `data`, which the third notebook uses, stays next to the notebooks. The Python packages are listed in `environment.yml`; with conda, run `conda env create -f environment.yml` and `conda activate homework-2`, then start `jupyter notebook` in that folder. Each notebook runs in less than a minute on a laptop.

We only expect to see the generated plots and your answers to the questions in the notebooks, included in your PDF. Each question in a notebook is followed by a code cell for your experiments and a text cell for your answers; when a question asks for several parameter values, write a loop or copy the relevant cell into that code cell, so that every requested figure appears in your PDF.

## Question 9: The Bayes-Kalman filter, dynamic programming and hidden Markov models (14 points)

Lecture 9. Notebook: `kalman_dp_hmm.ipynb`.
9.1 tracking a moving target **(3)**; 9.2 special cases **(2)**; 9.3 dynamic programming on a chain **(3)**; 9.4 the urn HMM **(3)**; 9.5 learning Markov models **(3)**.

## Question 10: Perceptrons, backpropagation and adversarial examples (14 points)

Lectures 10 and 11. Notebook: `perceptrons_backprop_adversarial.ipynb`.
10.1 the perceptron **(2)**; 10.2 logistic regression **(2)**; 10.3 a multi-layer perceptron and backpropagation **(3)**; 10.4 softmax regression (logistic regression with several classes) and a small convolutional network **(3)**; 10.5 adversarial examples **(4)**.

## Question 11: Attention and clustering (10 points)

Lectures 14 and 13 (k-means: Lecture 8). Notebook: `attention_clustering.ipynb`.
11.1 self-attention on an image **(3)**; 11.2 two normalisations **(3)**; 11.3 k-means cross-attention **(2)**; 11.4 a contrastive-learning toy **(2)**.